

Benchmarking Rule-Based, Learned, and Zero-Shot Vision-Language (GPT-4o) Grasp Prediction Under a Unified Evaluation Protocol

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Abstract—I compared three ways to predict a robotic grasp from an RGB image: a detector-assisted heuristic, a ResNet18 trained on grasp labels, and GPT-4o, a general-purpose vision-language model (VLM), used zero-shot. The protocol fixes the test images, input, output representation, and scoring rule on an object-wise split of the Cornell Grasping Dataset. It does not equalize supervision or tuning, so it compares complete pipelines. ResNet18 passed on 79.7% of test images (84.0% in a second, three-seed training round). The heuristic passed on 40.7% as frozen and on 57.7% after a fix prompted by test output; every statistical comparison uses the clean 40.7%. GPT-4o passed on 12.4% of calls across five independent runs per image. Under object-clustered resampling, ResNet18 led the heuristic by 39.0 points and the heuristic led one-call GPT-4o by 28.3. The errors differed. ResNet18 missed mostly on placement, while GPT-4o failed both criteria on 59.3% of calls and put its predicted center outside the hull of all labeled grasp corners on 53.4%. That pattern suggested GPT-4o might see a good grasp but fail to write it as coordinates. A fourth run tested this by showing the same model label-free candidate grasps drawn on the image and asking it to choose one by number. On three menus, one with every mark drawn the same size, its accuracy stayed at the menu’s random floor, and it preferred closing along the object’s long axis. Removing the coordinate output did not help, so the main failure is in choosing the grasp.

Index Terms—robotic grasping, computer vision, deep learning, vision-language models, multimedia data mining, spatial grounding

1. Introduction

A robot has to make a geometric decision before it can close its gripper: where the fingers should land, how the gripper should rotate, and how far the jaws should open. Someone building a first grasp predictor has three practical starting points. They can write the decision as rules, train a network on labeled grasps, or ask a hosted VLM, which needs no training data and almost no setup. The costs differ sharply, but published numbers do not say which to pick, because each family is usually tested in its own way. GR-ConvNet reports 96.6% object-wise on an extended Cornell

set with RGB-D input and a prediction at every pixel [4]. Single-rectangle regression reported 84.9% with RGB-D [3]. VLM grasping systems such as VLAD-Grasp are scored as whole pipelines in which the VLM is one stage among depth prediction, segmentation, and pose recovery [6]. Rule-based baselines are often described rather than scored. When input, output, split, and metric all change with the method, the gap between two papers mixes the method with the test.

I held those four things fixed and changed only where the grasp decision comes from. Every system saw the same 123 held-out RGB images of 35 objects, returned one grasp rectangle, and was scored by one implementation of the standard rectangle test [1], [2]. I reconstructed object identities so that no physical object crosses between training and test. “Unified” here means the same input, output, and metric. It does not mean equal footing: the systems differ in supervision, scaffolding, and development effort, and Table 1 lists those differences instead of hiding them. Each system is a complete pipeline of the kind a practitioner would build, so the results compare pipelines, not isolated reasoning abilities.

The paper asks two questions. First, under a fixed test, how much does the source of grasp knowledge (hand-written rules, task labels, or web-scale pretraining) change accuracy and where the errors fall? Second, when the VLM fails, is the failure in choosing a grasp or in writing it as pixel coordinates? Prior VLM pipelines sidestep the second question by design, since they rarely ask the model for final coordinates [5], [6]. I test it directly with a fourth run that keeps the model and images and removes the coordinate output.

The contribution is a measurement, not a new grasping method: a same-protocol comparison of three families on an object-wise split, with object-clustered statistics and the clean and post-hoc results kept apart; a shared failure taxonomy and a center-hull check that locate where each family fails; and a direct test of the most natural explanation for the VLM’s failure, which that explanation does not survive.

2. Related Work

Jiang *et al.* introduced the Cornell dataset and its rectangle representation of a grasp, which stores the center, orientation, opening, and jaw depth [1]. Under the usual

Cornell test, a prediction passes when it is within 30° of a labeled grasp and its intersection over union (IoU) is greater than 25% [2]. Redmon and Angelova regressed one rectangle per image from RGB-D and reported 84.9% object-wise [3], the published design closest to System B. GR-ConvNet’s dense pixel-wise output reaches higher scores [4].

VLM-based grasping systems usually do not ask the language model for the final pixel coordinates. Set-of-Mark prompting has the model choose among regions already marked on an image [5], and System D borrows that idea. VLAD-Grasp has a VLM draw the grasp, then recovers a pose with depth prediction, segmentation, and point-cloud alignment [6]. System C tests a smaller pipeline: show GPT-4o the image and ask it to return the coordinates itself. Its score describes that output method, not VLM-based grasping in general.

3. Experimental Design

3.1. Dataset, Split, and Metric

Cornell contains 885 RGB-D images of household objects, with several positive grasp rectangles on most images. I used only RGB. Cornell does not include object IDs, so I reconstructed them from the frame sequence. The procedure segmented the object from the photography platform and compared consecutive masks with color and area descriptors that tolerate rotation. I set the thresholds asymmetrically because the two possible mistakes have different costs. If two objects are merged, the combined group still stays in one split. If one object is split in two, its images can leak into both training and test.

Of the 884 frame boundaries, 251 (28.4%) fell in the ambiguous band. An automated first pass gave each one a decision and a confidence. In a blind 30-boundary audit, its calls agreed with my labels on 86.7% of confident cases but only 40.0% of uncertain ones, so its uncertain calls were not used. I decided 139 boundaries by hand, covering every uncertain call and every high-risk “different object” call, and accepted the other 112 confident same-object calls, where an error can only merge groups. Six boundaries stayed ambiguous after review. Four leaned toward “same object” and were merged. The other two had no lean, so instead of guessing I dropped the one image on the smaller side of each (pcd0435 and pcd0782). The open question in both cases was object identity, not the grasp annotations. That left 883 images in 234 groups, slightly under the 240 objects reported for this dataset [3], as expected from thresholds that err toward merging. With seed 42, training holds 164 objects (620 images), validation 35 (140), and test 35 (123). A second check confirmed that no object appears in more than one partition.

I scored each prediction against every positive rectangle on its image. A match to any one of them counts as correct. Before using the evaluator at scale, I checked hand-computed cases, rectangle round trips, and an independent rasterized IoU calculation. All predictions were converted back to the native 640×480 coordinate frame for the final comparison.

TABLE 1. WHAT EACH SYSTEM RECEIVES. TEST IMAGES, INPUT, OUTPUT, AND METRIC ARE IDENTICAL; EVERYTHING BELOW DIFFERS.

Sys.	Grasp labels	Scaffolding	Development	Test runs
A	3 constants	detector, mask	fixed table	2 (1 post-hoc)
B	620 train	none	3 nets, 2 rounds	2 (rules fixed)
C	none	none	1 prompt	1
D	A’s constants	mask, marks	1 prompt, 3 menus	3 (1 per menu)

3.2. System A: Detector-Assisted Heuristic

System A starts with a COCO-pretrained Faster R-CNN using a ResNet50-FPN-v2 backbone [7]. A fixed lookup table maps the detected category to a center, upper, or end grasp. I calibrated three global opening, jaw-size, and offset constants on training labels only. The COCO vocabulary misses most Cornell objects, so a platform-segmentation fallback supplies a box when the detector cannot. The rule closes the gripper across the box’s narrow dimension. It can output only 0° or 90° .

The frozen table scored 40.7% on its first and only clean test run. Its rendered predictions then exposed a defect: some accepted detections covered background clutter instead of the object on the platform. I added a guard requiring the detection to contain the segmented object’s centroid and recalibrated the constants on training data, and the score rose to 57.7%. Test output prompted that change, so 40.7% is System A’s result and is the value used in every statistical comparison. The 57.7% is reported only as a marked post-hoc figure.

3.3. System B: Learned Predictor

System B replaces ResNet18’s ImageNet classifier [8] with separate heads for center, orientation, and size. The orientation head predicts $(\cos 2\theta, \sin 2\theta)$. Doubling the angle handles the fact that a parallel-jaw grasp is unchanged by a 180° rotation. For images with several labels, I compute the loss against every rectangle and backpropagate only the best match. Averaging all valid labels could put the target between two good grasps, where neither one actually exists.

I trained a custom CNN, ResNet18, and ResNet34 on the 620 training images. Affine augmentation moved the rectangle corners with the pixels. Architecture, checkpoint, size-loss weight, and the custom CNN’s learning rate were chosen on validation data. ResNet18 had the best validation accuracy at 83.6%, so I froze that model before opening the test results.

That first round trained each architecture once, at one seed, with a constant learning rate, and kept the best of roughly a hundred noisy validation epochs. A second round kept the model, loss, data, and split and changed only the recipe: cosine learning-rate decay over 100 epochs, an exponential moving average of the weights, three seeds per architecture, and no checkpoint selection. Test-time augmentation over eight dihedral views, combined by taking the medoid rectangle, was adopted only because it raised mean validation accuracy. Four selection rules were written

down before the test split was opened a second time, and every configuration that touched test is reported.

3.4. System C: Zero-Shot VLM

System C sends the native RGB image to GPT-4o through OpenRouter [9]. It uses no grasp-specific training. The prompt asks for two fingertip coordinates and a jaw width in JSON, which together define the rectangle. I developed the prompt on 30 training images and froze it before test.

Each test image was sent in five independent, single-turn calls at the provider’s default temperature. I retried transport failures, but not replies that arrived and failed to parse. Retrying those would quietly turn a single-call baseline into a best-of-several system. Of 615 calls, 614 parsed. The primary score pools the five exchangeable runs to estimate one-call accuracy. Best-of-five is only a ceiling: it requires five calls and an oracle that knows when one of them is right.

3.5. System D: Marked-Candidate Selection

If GPT-4o can see where to grasp but cannot express it as coordinates, removing the coordinate output should recover its choice. System D shows the same model the same test images with candidate grasps drawn on a zoomed crop and asks for the number of the one it would use. Candidates come only from the platform-segmentation mask: principal-component analysis gives the centroid, long axis, and extent, and candidates are positions along the long axis crossed with four jaw-travel directions 45° apart, so every grasp angle is within the metric’s tolerance of some candidate. Opening and jaw width use System A’s train-calibrated constants. Numbering is re-shuffled on every call. Because the menu is fixed before the model sees it, each run carries its own floor (expected accuracy of a uniformly random choice) and ceiling (whether any candidate passes).

Three menus were run, each on the full test split with five repeats: twelve marks at three positions; four marks at the centroid only, drawn at real size; and the same four marks drawn at one uniform length, with the prompt stating that length is not to scale. The last two exist because on thin objects the correct marks are the smallest on the image, so a preference for large marks would mimic a preference for the long-axis grasp. The prompt was developed on 30 training images for parse rate and glyph reading only, and each menu’s test run was called once under its own sentinel.

3.6. Statistical and Failure Analysis

The 123 test images are repeated views of 35 unseen objects, not 123 independent samples. I formed the 95% confidence intervals from 20,000 object-clustered bootstrap resamples. Every view of an object, along with all five VLM calls, moves together. Paired differences use the same resamples and 20,000 object-level label-swap permutations.

The failure analysis gives each attempt one label: correct, angle only, overlap only, both criteria, or no prediction. I

TABLE 2. ACCURACY WITH OBJECT-CLUSTERED 95% CONFIDENCE INTERVALS. THE POST-HOC ROW (†) IS NOT USED IN ANY TEST.

System	Accuracy	95% CI
A: heuristic, as frozen	40.7%	[28.0, 53.2]
A: heuristic, post-hoc fix†	57.7%	[46.2, 68.2]
B: ResNet18, round 1	79.7%	[70.8, 87.1]
B: ResNet34, round 2, 3 seeds	84.0%	[73.9, 92.1]
C: GPT-4o, one call	12.4%	[8.0, 17.4]
C: best of five	35.0%	[24.6, 45.5]

also checked whether the predicted center falls inside the convex hull of all labeled grasp corners on the image. This is a loose spatial check, not a claim that the grasp would work on hardware.

4. Results

4.1. Overall Accuracy

ResNet18 passed 98 of 123 images (79.7%). The heuristic passed 50 as frozen (40.7%) and 71 after the post-hoc fix (57.7%). GPT-4o passed 76 of 615 calls (12.4%), and its five runs ranged from 9.8% to 14.6%. At least one call passed on 43 images, a best-of-five ceiling of 35.0%. Table 2 gives the object-clustered intervals.

Every paired test uses the clean heuristic. ResNet18 led it by 39.0 points [25.2, 52.9], with permutation $p < 0.0001$, and the heuristic led one-call GPT-4o by 28.3 points [16.9, 40.0], $p = 0.0001$. The heuristic’s lead over GPT-4o’s best-of-five ceiling, 5.7 points [−6.1, 18.0], $p = 0.45$, does not exclude zero. That comparison favored the heuristic by 22.8 points only after the post-hoc fix, so it is not a supported result. Giving each object equal weight produced 38.5%, 75.4%, and 10.5% for A, B, and one-call C, which changed the gaps but not their order.

The second training round chose ResNet34 on mean final-weight validation accuracy (81.4% against 81.2% for ResNet18). Its three seeds scored 84.6, 83.7, and 83.7% on test; ResNet18 under the same recipe averaged 82.6%. Re-running the first-round recipe at two new seeds gave 82.9% and 80.5%, so 79.7% was a low-side single draw, and the first round’s nine-point gap between ResNet18 and ResNet34 was run-to-run noise. Round 2 leads the clean heuristic by 43.4 points [29.9, 56.7], $p < 0.0001$; its 4.3-point lead over round 1 does not exclude zero. Splits and training data differ from [3], but System B’s RGB-only score is within 5.2 points of that RGB-D single-rectangle result in round 1, and 0.9 in round 2.

4.2. Failure Patterns

The error counts in Table 3 separate the systems more clearly than accuracy alone. The frozen heuristic most often failed both tests at once (34 of its 73 failures), and its predicted center fell outside the labeled-grasp hull on 28 of 115 predictions (24.3%), the mark of the clutter boxes the fix

TABLE 3. SHARED FAILURE TAXONOMY, COUNTS (% OF ATTEMPTS), OVER 123 IMAGES FOR A AND B AND 615 CALLS FOR C. A_0 IS THE FROZEN HEURISTIC; $A^\dagger$ IS POST-HOC.

Outcome	A_0	$A^\dagger$	B	C
Correct	50 (40.7)	71 (57.7)	98 (79.7)	76 (12.4)
Angle only	15 (12.2)	13 (10.6)	4 (3.3)	85 (13.8)
Overlap only	16 (13.0)	18 (14.6)	15 (12.2)	88 (14.3)
Both	34 (27.6)	13 (10.6)	6 (4.9)	365 (59.3)
No prediction	8 (6.5)	8 (6.5)	0 (0.0)	1 (0.2)

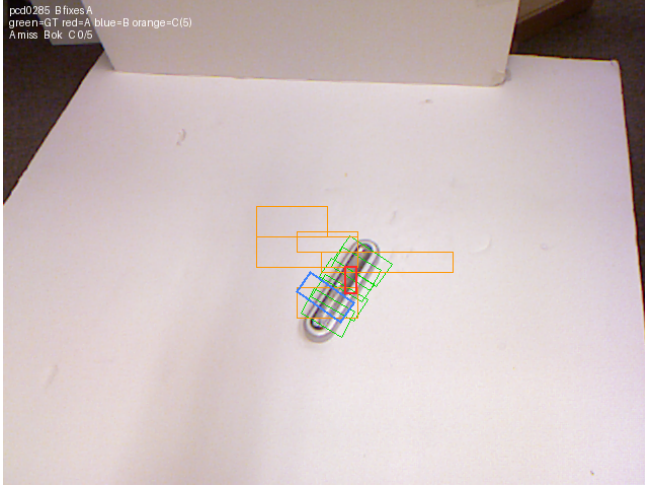


Figure 1. One test image under the shared coordinate system. Labeled grasps are green, System A red, System B blue, and System C’s five repeats orange.

later rejected. After the fix, 13 of A’s 52 failures missed only the angle test. That is partly built in, since A can output only 0° or 90° , but the limit is loose: only 5 test images (4.1%) have no labeled grasp within the metric’s 30° of either axis, and 103 (83.7%) have one within 15° . On the 20 images whose labeled grasps all lie more than 15° from both axes, corrected A passed 40.0%, against 61.2% on the rest. B had four angle-only failures; 15 of its 25 misses failed only on overlap, so placement was its larger problem. C failed both tests on 365 of 615 calls (59.3%).

The center-hull check shows where C’s failures begin. Corrected A landed outside the hull on 8 of 115 predictions (7.0%) and B on 8 of 123 (6.5%); C did so on 328 of 614 parsed calls (53.4%). Figure 1 shows the shared coordinate frame on one image. This is not an even contest by itself: A starts from a detected or segmented object, B was trained on Cornell, and C can place its coordinates anywhere. I use the check to locate a stage of failure, not to estimate a shared capability.

The five GPT-4o calls also disagreed with one another. Mean pairwise rectangle agreement was 22.0%, and mean pairwise IoU was 0.13. All five calls failed on 80 of 123 images. All five passed on one image. The remaining 42 images were unstable: identical pixels sometimes produced a passing grasp and sometimes did not.

TABLE 4. SYSTEM D ON THREE CANDIDATE MENUS, SAME MODEL AND TEST IMAGES, FIVE REPEATS EACH. “LONG AXIS” IS THE SHARE OF CALLS CHOOSING THE ORIENTATION WHOSE JAWS TRAVEL ALONG THE OBJECT’S LONG AXIS; CHANCE IS 25% ON THE FOUR-MARK MENU.

Menu	Marks	Acc.	Floor	Ceil.	Long axis
Full, 3 positions	12	18.7	20.1	79.7	70%
Centroid, real	4	26.5	24.6	65.9	41%
Centroid, uniform	4	22.3	24.6	65.9	45%

4.3. System D

Table 4 gives the three menus. On none of them does the model’s one-call accuracy differ from a random choice among the same candidates (paired object-clustered differences -1.4 $[-6.6, +4.0]$, $+1.9$ $[-2.2, +6.3]$, and -2.3 $[-6.3, +1.7]$), while the ceilings are 79.7% and 65.9%. Its choices are not random, though. On the full menu it chose the candidate whose jaws travel along the object’s long axis near one end in 61% of calls, a grasp type that passes 17% of the time, with reasoning such as “grips the narrow top edge near the center of mass”. With four marks at the centroid drawn at one uniform size, it still chose the long-axis direction in 45% of calls against a 25% chance rate, and the short-axis candidate, which passes on 55% of images, in 17%. Stated confidence was “high” on 556 of 560 full-menu calls, 21% of them correct. Every reply parsed. The centroid candidate closing across the short axis, with no model at all, scored 55.3% [41.8, 68.0], level with the corrected heuristic. It was defined after the main runs and reuses A’s post-fix constants, so it belongs with the post-hoc figures.

5. Discussion

5.1. Why GPT-4o Fails

Before System D, three explanations fit C’s results. The model might pick a sensible region in words and lose it while writing pixels (coordinate binding); it might misread the image’s coordinate frame or scale; or it might choose a poor grasp in the first place. The informal evidence for coordinate binding was that the rationales I read on a sample of C’s failures named a plausible part of the object while the coordinates landed elsewhere, which was not a blinded study. The first two explanations both blame the coordinate output, so both predict that removing it would move accuracy toward the menu’s 79.7% ceiling. System D removed coordinates and scale altogether, and accuracy stayed at the random floor on all three menus. The preference for closing along the long axis survived a menu with no end-straddling candidates and one with every mark the same size. Coordinate errors may still add to C’s misses (D minus C was $+6.3$ $[-1.6, +15.0]$, $p = 0.19$), but they cannot account for them. From a top-down photograph, two fingertip pads straddling the end of an elongated object do look like they pinch a narrow part; in three dimensions the second finger lands on top of the object. The evidence fits a failure of grasp reasoning

from a single 2-D view better than a failure to bind text to coordinates.

5.2. Practical Implications

For someone choosing a starting point, three conclusions follow on this benchmark. With a few hundred labeled images, fine-tuning a small pretrained CNN was the strongest option by a wide margin, and its remaining errors were placement errors, which points to where more data or a dense output would help. Without labels, the object mask carried most of the useful signal: the COCO detector missed most Cornell objects and, as frozen, often boxed background clutter, while the label-free short-axis rule, a post-hoc baseline, scored 55.3%. Zero-shot GPT-4o should not be the source of grasp geometry, whether asked for coordinates or for a choice among marks. Its stated confidence, “high” on 556 of 560 System D calls, cannot filter bad grasps, and repeated calls on one image agreed on only 22% of rectangles. A VLM in a grasping system should leave the geometric decision to another component, as VLAD-Grasp does with depth and point-cloud alignment [6]. For evaluation practice, the 17-point move from one test-prompted fix shows why the clean number should carry the inference.

5.3. Limitations

The evaluation is narrow: one object-wise split of one small benchmark, RGB without depth, one VLM with one prompt per output format, and no robot trials. The 123 test images come from only 35 objects, so the intervals are wide, and a fresh split, object-wise cross-validation, or a second dataset is needed to confirm the ranking. Cornell’s annotations are not exhaustive, so a rectangle that fails the metric might still work on a real object. The development budgets differ (Table 1). B received three architectures, validation tuning, and a second training round, against one frozen prompt for C and one per menu for D. The test split was opened twice for B under rules fixed beforehand, and twice for A, whose second result is post-hoc. System D used one glyph style and one candidate generator. An early formatting check on about 550 grasp rectangles probably included a few images that later entered test, although no parameter or rule came from that check.

6. Conclusion

Under one fixed input, output, and metric, ResNet18 scored 79.7% (84.0% over three seeds in a second round), the frozen heuristic 40.7% (57.7% after a post-hoc fix), and GPT-4o 12.4% per call. Their error locations add context to that ranking. The frozen heuristic often acted on background clutter, ResNet18 usually got the angle and missed the placement, and GPT-4o often failed before either distinction mattered. Given marked candidate grasps instead of a coordinate output, the same model chose no better than chance on three menus and kept preferring a grasp along

the object’s long axis even with every mark drawn the same size. On this benchmark the problem is the grasp choice, not only its expression. Whether depth input, or a model with a stronger three-dimensional prior, changes that choice is the next question.

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Author’s work. I am the sole author. I chose the research question and designed the study: the protocol, the split policy, all four systems, and the statistical analysis. I labeled the blind boundary audit, decided the 139 hand-reviewed boundaries, ran the experiments, inspected their outputs, and checked every number and citation in this paper against the saved results.

AI assistance. Anthropic Claude Code, an AI coding assistant, helped with code development throughout, made the first-pass calls on the 251 ambiguous boundaries before my audit, implemented the second training round and System D to my design, and helped revise this camera-ready version, including re-running System A’s clean statistics. OpenAI Codex helped shorten and edit an earlier version of the text. GPT-4o appears only as the model under test in Systems C and D.

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